

BUSINESS CASE	
Proposed Project	Smart Parking Billing System (SPBS)
Date Produced	September 14, 2026
Background	Campus parking system utilizes various distinct parking zones, including underground parkades, standard surface lots, economy outer lots, and block-heater electrical plug equipped lots. Currently, physical terminal meters are location-bound. People who do not use a mobile app and need to extend their parking duration have to physically walk back to the specific terminal meter tied to their parking lot to pay, causing significant inconvenience - especially during harsh weathers or while attending class. This project focuses on building a Java terminal application using Model-View-Controller (MVC) architecture to allow cross-zone billing and systematically validate its logic using JUnit test suites.
Business Need/ Opportunity	Cross-Zone Ticket Extensions: Allow drivers to enter their vehicle license plate at <i>any</i> terminal, select their active parking lot zone, and purchase or extend parking time remotely without walking back to their original parking lot. Variable Zone Pricing Logic: Automate tiered fee structures based on zone type (e.g., higher rates for underground or plug-in stalls vs. lower rates for far economy lots) and duration limits. Software Validation & Testing Sandbox: Create a clear set of business rules and state changes to demonstrate white-box (path/data flow) and black-box (boundary value, decision table, state transition) unit testing using JUnit.
Options	Option 1: Do Nothing (Maintain location-bound meters): Continue using specific lot physical meters that require drivers to walk back to their specific lot to extend time. Option 2: Proceed with a Smart Parking Billing System: Develop a Java MVC application that handles cross-zone ticket management, tiered pricing, and automated status updates with full JUnit test coverage.
Cost-Benefit Analysis	
OPTION 1 - Do Nothing Costs: Driver inconvenience and wasted time traveling back to lot-specific meters. Increased risk of expired tickets and fines due to distance barriers.	

Lack of flexible, centralized zone management.

Benefits:

No software design, setup, or development time required.

OPTION 2 - Universal Smart Parking Billing System

Costs:

Time and effort spent across the term designing, implementing, and documenting the Java software.

Development effort required to write JUnit test suites for multi-zone calculations, boundary conditions, and state changes.

Benefits:

High driver convenience by allowing time extensions from any terminal.

Fast, automated fee calculation with zero manual billing errors across different zones.

Fulfills all requirements for MVC architecture and test-driven development.

Recommendation

Proceed with **Option 2**. Solving the physical meter constraint provides a practical, well-scoped engineering problem that maps directly to the required architecture and test suites.